Dummit and Foote Algebra exam practice problems and solutions (questions posed by ChatGPT)

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1. What is a group? Give the formal definition and give one example each of an abelian group, nonabelian group, finite group, and infinite group.

A Group $(G, \cdot)$ is a set G equipped with a binary operation

$$\cdot : G \times G \to G$$

satisfying the following axioms:

Closure. For all $g, h \in G$,

$$g \cdot h \in G.$$

Associativity. For all $g, h, k \in G$,

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

Identity. There exists an element $e \in G$ such that, for every $g \in G$,

$$e \cdot g = g = g \cdot e.$$

Inverses. For every $g \in G$, there exists an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = e = g^{-1} \cdot g.$$

Examples:

- **Abelian group:** $(\mathbb{Z}, +)$.
- **Nonabelian group:** The symmetric group S_3 under composition.
- **Finite group:** $(\mathbb{Z}/3\mathbb{Z}, +)$.
- **Infinite group:** $(\mathbb{R}, +)$.

2. What is a subgroup? State and prove the subgroup criterion. A subgroup H is a non-empty subset of G that is also a group. H is a subgroup of G if and only if for all $a, b \in H$ then $ab^{-1} \in H$

Proof: " $\implies$ "

Suppose H is a subgroup in G. Then for $a, b \in H$ by the inverse property $b^{-1} \in H$. Next by the closure property $ab^{-1} \in G$.

Proof: " $\impliedby$ "

Suppose for all $a, b \in H$, then $ab^{-1} \in H$.

Identity. $e \in H$

Since H is nonempty, choose $a = x$ and $b = x$. Thus, we have $xx^{-1} = e \in H$. Hence H has an identity. Next, since $e \in H$ and $x \in H$

Inverses. $x \in H \implies x^{-1} \in H$

Since $a \in H$, choose $a = e$ and $x \in H$. Thus, we have $ex^{-1} = x^{-1} \in H$

Closure. For all $x, y \in H$, $xy \in H$

Let $x, y \in H$. Since $y^{-1} \in H$ choose $a = x$ and $b = y^{-1}$. Thus, we know $x(y^{-1})^{-1} = xy \in H$

Associativity. Follows naturally from G because all elements in H are also in G.

3. Compute orders. Find the order of every element in S_3, D_8, $\mathbb{Z}/12\mathbb{Z}$, and $(\mathbb{Z}/12\mathbb{Z})^\times$.

All elements in S_n can be written as a product of independent cycles (ex: $(1\,4\,5)(3\,2) \in S_5$) whose order is the LCM of the length of each cycles.

Elements of D_n are generated by elements r and f such that

$$D_8 = \langle r, f : r^4 = e,\ f^2 = e,\ fr = r^3 f \rangle$$

Elements of $\mathbb{Z}/12\mathbb{Z}$ are the integers mod 12. Thus, the order of every element x is the smallest positive integer a such that $a \cdot x \equiv 0 \mod 12$

Finally $(\mathbb{Z}/12\mathbb{Z})^\times$ are all integers less than 12 with a multiplicative inverse mod 12. Or in other words

$$(\mathbb{Z}/12\mathbb{Z})^\times = \{x \in \mathbb{Z}/12\mathbb{Z} : \gcd(x, 12) = 1\} = \{1, 5, 7, 11\}$$

Thus, the order of every element x is the smallest positive integer a such that $x^a \equiv 1 \mod 12$

	Element:	**Power**	**Order**
	e	$e^1 = e$	1
	$(1\,2)$	$(1\,2)^2 = e$	2
S_3:	$(2\,3)$	$(2\,3)^2 = e$	2
	$(1\,3)$	$(1\,3)^2 = e$	2
	$(1\,2\,3)$	$(1\,2\,3)^3 = e$	3
	$(1\,3\,2)$	$(1\,3\,2)^3 = e$	3

	Element:	**Power**	**Order**
	e	$e^1 = e$	1
	r	$r^4 = e$	4
	r^2	$(r^2)^2 = r^4 = e$	2
D_8:	r^3	$(r^3)^4 = (r^4)^3 = e$	4
	f	$f^2 = e$	2
	rf	$(rf)^2 = rfrf = rr^3ff = r^4f^2 = e$	2
	r^2f	$(r^2f)^2 = r^2fr^2f = r^2r^6ff = r^8f^2 = e$	2
	r^3f	$(r^3f)^2 = r^3fr^3f = r^3r^9ff = r^{12}f^2 = e$	2

	Element:	**Power**	**Order**
	0	$1 \cdot 0 = 0 \equiv 0 \mod 12$	1
	1	$12 \cdot 1 = 12 \equiv 0 \mod 12$	12
	2	$6 \cdot 2 = 12 \equiv 0 \mod 12$	6
	3	$4 \cdot 3 = 12 \equiv 0 \mod 12$	4
	4	$3 \cdot 4 = 12 \equiv 0 \mod 12$	3
$\mathbb{Z}/12\mathbb{Z}$:	5	$12 \cdot 5 = 60 \equiv 0 \mod 12$	12
	6	$2 \cdot 6 = 12 \equiv 0 \mod 12$	2
	7	$12 \cdot 7 = 84 \equiv 0 \mod 12$	12
	8	$3 \cdot 8 = 24 \equiv 0 \mod 12$	3
	9	$4 \cdot 9 = 36 \equiv 0 \mod 12$	4
	10	$6 \cdot 10 = 60 \equiv 0 \mod 12$	6
	11	$12 \cdot 11 = 132 \equiv 0 \mod 12$	12

	Element:	**Power**	**Order**
	1	$1^1 = 1 \equiv 1 \mod 12$	1
$(\mathbb{Z}/12\mathbb{Z})^\times$:	5	$5^2 = 25 \equiv 1 \mod 12$	2
	7	$7^2 = 49 \equiv 1 \mod 12$	2
	11	$11^2 = 121 \equiv 1 \mod 12$	2

4. Prove that every subgroup of a cyclic group is cyclic.

Let G be a cyclic group and H a subgroup of G generated by x. All elements of H are also powers of x. Choose the smallest positive integer k such that $x^k \in H$. Thus, all powers of x^k are also in H. Thus, it would suffice to show that x^k generates H. Suppose x^k does not generate H. Thus, there must be $g \in H$ such that

$$g = x^{km+r} = x^{km}x^r$$

Such that $r < k$. But since $g \in H$ and $x^{km} \in H$ then by the group operation $x^r \in H$. But we already chose k to be the smallest positive integer. Thus, $r = 0$. Hence x^k generates the whole group H and is cyclic.

5. Generators of cyclic groups. Determine all generators of $\mathbb{Z}/n\mathbb{Z}$. Prove your answer.

It would suffice to show that g^k generates $G = \mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(k, n) = 1$.

Proof: " $\Longrightarrow$ "
Suppose G is generated by g and let g^k also be a generator of G. Thus, there exists m such that $g = (g^k)^m$ and thus

$$km \equiv 1 \mod n$$

Rewriting, we see there must exist an integer q such that

$$k(m) - n(q) = 1$$

By Bézout's identity, if a linear combination of two integers n, k equals one then

$$\gcd(n, k) = 1$$

Proof: " $\Longleftarrow$ "

Suppose $\gcd(n, k) = 1$. Then consider the group H generated by g^k where $g^n = e$. Since all elements of H are orders of g, then $H \subset G$. Next, since $\gcd(n, k) = 1$, for any positive integer m, where $m < n$ then

$$qk \equiv m \mod n$$

Thus, $g^m = g^{qk} \in H$ Thus, we see that $G \subset H$ and Thus, $H = G$ and hence g^k generates G.

Thus, all generators of $\mathbb{Z}/n\mathbb{Z}$ are elements $k < n$ such that $\gcd(k, n) = 1$

6. Cosets and Lagrange. Define left and right cosets. Prove Lagrange's theorem.

Let G be a group and H a subgroup of G
Left Coset: A set $gH = \{gh : h \in H\}$

Right Coset: A set $Hg = \{hg : h \in H\}$

Lagrange's theorem: Let G and H be finite groups, if H is a subgroup of G, then $|H|$ divides $|G|$.

Suppose that H is a subgroup of G. Then, for all $g \in G$, we get gH is a subset of G. let $g_0 = e$ then our first coset is $g_0H = H$. Next choose $g_1 \notin g_0H$. We get a new coset g_1H. We keep choosing elements such that

$$g_n \notin \bigcup_{i=0}^{n-1} g_iH$$

Since G is a finite group, this algorithm must terminate. Thus, we now have the set of all cosets

$$\left\{g_iH : g \in G, g_n \neq g_m, \bigcup g_iH = G\right\}$$

Next we want to show that these cosets are disjoint, and they are all equal size. Suppose $g_n \neq g_m$ where $g_m H \cap g_n H \neq \varnothing$, then there must exist $h_1, h5_2 \in H$ such that $g_m h_1 = g_n h_2$. Next we see

$$\begin{aligned} g_m h_1 = g_n h_2 &\implies g_m = g_n h_2 h_1^{-1} \\ &\implies g_m \in g_n H \end{aligned}$$

But we chose $g_n \notin g_m H$

Thus, we get a contradiction and $g_m H \cap g_n H = \varnothing$. Next suppose that there exists a coset such that $|gH| < |H|$. Meaning that there are duplicate elements in gH. Thus, there must exists $h_1, h_2 \in H$ such that $h_1 \neq h_2$, but $gh_1 = gh_2$. But this means that $h_1 = h_2$. A contradiction. Thus, for all cosets, $|g_i H| = |H|$. Since The disjoint union of these n cosets is G and they all have order $|H|$, then $n|H| = |G|$ Thus, $|H|$ divides $|G|$.

7. Normal subgroups. What does it mean for $N \trianglelefteq G$? Prove that $N \trianglelefteq G$ if and only if the left cosets and right cosets of N are the same.

Let N be a subgroup of G. N is a normal subgroup if and only if for all $g \in G$ then $gNg^{-1} = N$.

Proof: " $\implies$ "
Suppose $N \trianglelefteq G$ and fix $g \in G$, then $g \in G$, $gNg^{-1} = N$. Multiplying on the right by g we see that

$$gN = Ng$$

Proof: " $\impliedby$ "
Suppose the left and right cosets are equal. Then for sone $n_1, n_2 \in N$, we have

$$gn_1 = n_2 g$$

Multiplying on the right by g^{-1} we get

$$\begin{aligned} gn_1 g^{-1} = n_2 &\implies gn_1 g^{-1} \in N \\ &\implies gNg^{-1} = N \end{aligned}$$

For brevity later on, we want to show that for all $n \in N$ and $g \in G$

$$gng^{-1} \in N \implies N \trianglelefteq G$$

Suppose that $gn_1 g^{-1} \in N$. Thus

$$gNg^{-1} \subseteq N$$

Multiplying on the left and right by g^{-1} and g respectively

$$N \subseteq g^{-1} N g$$

But we also know that the first equation applies to all g and g^{-1}, by symmetry

$$g^{-1} N g \subseteq N$$

Thus

$$gNg^{-1} = N$$

8. Quotient groups. Define G/N. Prove that the operation $(aN)(bN) = abN$ is well-defined if and only if $N \trianglelefteq G$.

Let $N \trianglelefteq G$. The quotient group G/N is the group of all left cosets of N. That being

$$G/N = \{gN : g \in G\}$$

With binary operation $(aN)(bN) = abN$.
Proof: " $\implies$ "

Suppose that $(aN)(bN) = abN$ is well defined. Then if $aN = a'N$ and $bN = b'N$ then $abN = a'b'N$. Choose $a = g$, $b = g^{-1}$, $a' = gn$ and $b' = g^{-1}$. We know $gnN = gN$ since $nN = N$

$$\begin{aligned}
(aN)(bN) = abN &\implies (gN)(g^{-1}N) = gg^{-1}N = N && \text{by our substitution} \\
(a'N)(b'N) = a'b'N &\implies (gnN)(g^{-1}N) = gng^{-1}N && \text{by our substitution} \\
&\implies gng^{-1}N = N && \text{since the first and second equations are equal}
\end{aligned}$$

Thus, $gng^{-1} \in N$. Hence $N \trianglelefteq G$.
Proof: " $\impliedby$ "

Suppose $N \trianglelefteq G$. Let $a, a', b, b' \in G$ such that $aN = a'N$ and $bN = b'N$. Thus, we can choose a' such that $a' = an_1$ and b' such that $b' = bn_2$ where $n_1, n_2 \in N$. Thus, we also get that $a'b' = an_1bn_2$. Using normality of N (being $gN = Ng$) Then there exists n_3 such that $an_1bn_2 = abn_3$ Thus, $a'b' \in abN$. Hence

$$a'b'N = abN$$

Thus, the operation is well defined.

9. Homomorphisms. Define kernel and image. Prove that $\ker\varphi \trianglelefteq G$ for any group homomorphism $\varphi : G \to H$

The kernel of a homomorphism φ is defined as

$$\ker(\varphi) = \{g \in G : \varphi(g) = e_H\}$$

The Image of a homomorphism is defined as

$$\mathrm{Im}(\varphi) = \{\varphi(g) \in H : g \in G\}$$

First fix $g \in G$ and $k \in \ker(\varphi)$. Evaluate $\varphi(gkg^{-1})$

$$\begin{aligned}
\varphi(gkg^{-1}) &= \varphi(g) \cdot \varphi(k) \cdot \varphi^{-1}(g) && \text{by properties of the homomorphism} \\
&= \varphi(g) \cdot \varphi^{-1}(g) && \text{By definition of } k \in \ker\varphi \\
&= e_H && \text{Since } \varphi(g) \text{ and } \varphi^{-1}(g) \text{ are inverses}
\end{aligned}$$

Since $\varphi(gkg^{-1}) = e_H$ then $gkg^{-1} \in \ker(\varphi)$. By the previous problem

$$gkg^{-1} \in \ker\varphi \implies \ker\varphi \trianglelefteq G$$

10. First isomorphism theorem. State and prove the first isomorphism theorem for groups

The First isomorphism theorem for groups states that for any homomorphism $\varphi : G \to H$, then

$$G/\ker\varphi \cong \varphi(G)$$

Shorten $\ker\varphi = K$ for brevity. Consider the function $\psi : G/K \to \varphi(G)$ such that

$$\psi(gK) = \varphi(g)$$

Suppose $gK = hK$.

$$\begin{aligned}
gK = hK &\implies h^{-1}gK = K && \text{Multiplying on the left by } h^{-1}\\
&\implies h^{-1}g \in K && \text{since } K \trianglelefteq G\\
&\implies \varphi(h^{-1}g) = e_H && \text{Since we defined } K = \{g \in G : \varphi(g) = e_H\}\\
&\implies \varphi(g) = \varphi(h) && \text{By properties of the homomorphism}
\end{aligned}$$

Thus, ψ is well defined. Next suppose $\psi(gK) = \psi(hK)$ then

$$\begin{aligned}
\varphi(g) = \varphi(h) &\implies \varphi(h^{-1}g) = e_h && \text{by homomorphism properties}\\
&\implies h^{-1}g \in K && \text{by definition of K}\\
&\implies gK = hK && \text{by coset properties}
\end{aligned}$$

Thus, ψ is surjective. Next evaluate $\psi(gK)\psi(hK)$

$$\begin{aligned}
\psi(gK)\psi(hK) &= \varphi(g)\varphi(h)\\
&= \varphi(gh)\\
&= \psi(ghK)
\end{aligned}$$

Meaning ψ is a homomorphism. Finally evaluate $\ker\psi$. We want to find all $gK \in G/K$ such that $\psi(gK) = e_H$.

$$\psi(gK) = \varphi(g) = e_H \iff g \in K$$

Thus, the only element in $\ker\psi$ is when $g = e_G$, and hence $eK = K$. Meaning that $\ker\psi$ is trivial in G/K and hence ψ is injective. Since ψ is injective, surjective, and a homomorphism. Then ψ is an isomorphism. Thus

$$G/K \cong \varphi(G)$$

11. Permutation groups. Write the following permutation as a product of disjoint cycles, then compute its order

$$\sigma = (1\,4\,7)(2\,5)(3\,8\,6\,9)(4\,2\,3)$$

Since the highest number is 9 then $\sigma \in S_9$. Thus, we check each number 1 through 9
$\sigma(1) = 4$, $\sigma(2) = 8$, $\sigma(3) = 7$, $\sigma(4) = 5$, $\sigma(5) = 2$, $\sigma(6) = 9$, $\sigma(7) = 1$, $\sigma(8) = 6$, $\sigma(9) = 3$
Next, starting from 1 (arbitrary decision), we then append $\sigma(1)$ then $\sigma(\sigma(1)$ and continue until we get back to 1. If there are any remaining numbers choose one arbitrarily n then we take $\sigma(n)$, append $\sigma(\sigma(n))$ and continue until getting back to n. Continue until all numbers are mapped. For this example we get

$$\sigma = (1\,4\,5\,2\,8\,6\,9\,3\,7)$$

The order of any element is just the lcm of all the orders of the disjoint cycles. Since there is only one cycle then the order is the length of the string. Thus, $|\sigma| = 9$

12. Alternating groups. Define A_n. Prove that $A_n \trianglelefteq S_n$ and compute $[S_n : A_n]$

Recall the definition of S_n

$$S_n = \{\sigma : \{1, 2, \cdots, n\} \to \{1, 2, \cdots, n\} \text{ where } \sigma \text{ is bijective}\}$$

Before doing that I want to show that all elements of S_n can be decomposed into products of 2-cycles. First suppose that σ is just one k cycle. Then using a common identity, we can split it into a product of 2-cycles as shown

$$\sigma = (a_1\, a_2\, \cdots\, a_k) = (a_1\, a_k)(a_1,\, a_{k-1}) \cdots (a_1\, a_3)(a_1\, a_2)$$

Since this identity applies to any length of cycles, then it would also apply to products of disjoint cycles. Thus, all elements can be written as a product of 2-cycles.
We define $A_n \subset S_n$ such that

$$A_n = \{\sigma \in S_n : \sigma \text{ is composed of an even number of 2-cycles}\}$$

Define a parity function $p : S_n \to \mathbb{Z}/2\mathbb{Z}$ such that if $\sigma \in S_n$ is a product of k amount of 2-cycles, then

$$p(\sigma) = k \mod 2$$

Trivially it's well defined since parity is invariant for any two equal cycles. It's also a homomorphism since composition of two elements if S_n just adds the amount of 2-cycles. Find the $\ker p$. $\sigma \in \ker(p)$ iff it can be written as an even number of 2-cycles. Thus, $A_n = \ker p$ and $A_n \trianglelefteq S_n$. Next recall that $[S_n : A_n] = |S_n/A_n|$. By the first isomorphism theorem

$$S_n/A_n \cong p(S_n) \cong \mathbb{Z}/2\mathbb{Z}$$

Thus

$$[S_n : A_n] = |S_n/A_n| = |\mathbb{Z}/2\mathbb{Z}| = 2$$

13. Group actions. Define a group action. Given an action of G on a set X, define the orbit and stabilizer of $x \in X$, then prove the orbit-stabilizer theorem.

Let G be a group and X be a set. A group action is a binary operation $G \cdot X \to X$ such that for $g, h, e \in G$ and $x \in X$, then $ex = x$ and $g(hx) = (gh)x$. If we fix $x \in X$, the orbit of x is

$$\text{Orb}(x) = \{g \cdot x : g \in G\}$$

The stabilizers of x is

$$\text{Stab}(x) = \{g \in G : g \cdot x = x\}$$

We want to first show that $\text{Stab}(x)$ is a subgroup of G. Choose g and h in $\text{Stab}(x)$. Then evaluate $(gh^{-1}) \cdot x$. Since $h \cdot x = x$ then left multiplying by h^{-1}, we get $x = h^{-1} \cdot x$ Thus, $h^{-1} \in \text{Stab}(x)$. So $(gh^{-1}) \cdot x$

$$\begin{aligned}(gh^{-1}) \cdot x &= g(h^{-1}x) \\ &= g \cdot x \\ &= x\end{aligned}$$

Thus, $gh^{-1} \in \text{Stab}(x)$. By the subgroup criterion, $\text{Stab}(x)$ is a subgroup of G.

Next, fix $x \in X$. Define a function $\varphi : G \to \text{Orb(x)}$ such that

$$\varphi(g) = g \cdot x$$

φ is trivially surjective since all of G acts on all of X by definition of a group action. Next suppose that $\varphi(g) = \varphi(h)$ then

$$\begin{aligned}\varphi(g) = \varphi(h) &\implies gx = hx \\ &\implies h^{-1}gx = x \\ &\implies h^{-1}g \in \text{Stab}(x)\end{aligned}$$

Thus, $g \equiv h \mod \text{Stab(x)}$ Meaning there exists another bijection $\psi : G/\text{Stab}(x) \to \text{Orb}(x)$ defined by

$$\psi(g \text{ Stab}(x)) = g \cdot x$$

Thus, since a bijection exists, $|G/\text{Stab}(x)| = |\text{Orb}(x)|$ and thus

$$|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|$$

14. Class equation. State and prove the class equation for a finite group acting on itself by conjugation.

First consider a group G acting on a set X by $g \cdot x$. We want to show that the orbits partition the set into disjoint unions. Let $x \in X$ be arbitrary. Then $e \in G$ acts on x by $x \cdot e$. Thus, $x \in \text{Orb}(x)$. Thus, every element of X belongs in an orbit. Next suppose given the orbits of x_1 and x_2 we have

$$\text{Orb}(x_1) \cap \text{Orb}(x_2) \neq \varnothing$$

Then there exist $g_1, g_2 \in G$ such that

$$g_1 \cdot x_1 = g_2 \cdot x_2$$

Left multiplying by g_1^{-1}

$$x_1 = g_1^{-1}(g_2 \cdot x_2)$$

Thus

$$\text{Orb}(x_1) \subseteq \text{Orb}(x_2)$$

But by reversing the argument

$$\text{Orb}(x_2) \subseteq \text{Orb}(x_1)$$

Thus

$$\text{Orb}(x_1) = \text{Orb}(x_2)$$

Hence no two orbits can partially overlap. Thus, for any group G acting on any set X. We can consider all $x \in X$ and only choose distinct orbits. Thus, choose $x_1, x_2, \cdots, x_n \in X$ with distinct orbits such that.

$$X = \text{Orb}(x_1) \sqcup \text{Orb}(x_2) \sqcup \cdots \sqcup \text{Orb}(x_n)$$

Thus

$$|X| = |\text{Orb}(x_1)| + |\text{Orb}(x_2)| + \cdots + |\text{Orb}(x_n)|$$

Which is called the Partition Equation for a group action.

Finally consider the the group G acting on itself by conjugation. That being $G \cdot G \to G$ is defined by

$$g \cdot x = gxg^{-1}$$

Consider some $x \in G$. The conjugacy class of $x \in G$ is defined by the set of all conjugations of x with elements of G.

$$\text{Cl}(x) = \{gxg^{-1} : g \in G\}$$

The center of a group $Z(G)$ is defined as the set of "Abelian elements" loosely speaking. Formally as

$$Z(G) = \{g \in G : gx = xg, \forall x \in G\}$$

Finally, the centralizer of an element $C(x)$ are the elements that commute with x

$$\text{C}(x) = \{g \in G : gx = xg\}$$

Fix $x \in G$ and find the orbit and stabilizers of x.

$$\begin{aligned} \text{Orb}(x) = \{g \cdot x : x \in G\} &\implies \text{Orb}(x) = \{gxg^{-1} : g \in G\} \\ &\implies \text{Cl}(x) = \text{Orb}(x) \end{aligned}$$

$$\begin{aligned} \text{Stab}(x) = \{g \in G : g \cdot x = x\} &\implies \text{Stab}(x) = \{g \in G : g \cdot x = x\} \\ &\implies \text{Stab}(x) = \{g \in G : gxg^{-1} = x\} \\ gxg^{-1} = x &\implies gx = xg \\ &\implies \text{Stab}(x) = C_G(x) \end{aligned}$$

Recall the orbit stabilizer theorem for this case that for some fixed $x \in G$

$$|\text{Orb}(x)| = [G : \text{Stab}(x)] = \frac{|G|}{|\text{Stab}(x)|}$$

Substituting we get

$$|\text{Cl}(x)| = [G : C(x)]$$

Since $\text{Cl}(x) = \text{Orb}(x)$, substitution into the partition equation we have (we label elements y_i for re-indexing later)

$$G = \text{Cl}(y_1) \sqcup \text{Cl}(y_2) \sqcup \cdots \sqcup \text{Cl}(y_m)$$

Consider all the conjugacy classes with one element

$$\begin{aligned} |\text{Cl}(x)| = 1 &\implies gxg^{-1} = x, \forall g \in G \\ &\implies gx = xg, \forall g \in G \\ &\implies x \in Z(G) \\ |\text{Cl}(x)| = 1 &\iff x \in Z(G) \end{aligned}$$

Thus, for $|\text{Cl}(x_i)| = 1$

$$|\text{Cl}(x_i)| + |\text{Cl}(x_j)| + \cdots = |Z(G)|$$

Finally factoring out $Z(G)$ in our partition equation

$$|G| = |Z(G)| + |\text{Cl}(x_1)| + |\text{Cl}(x_2)| + \cdots + |\text{Cl}(x_n)|$$

$$|G| = |Z(G)| + \sum_{i=i}^{n} |\text{Cl}(x_i)|$$

Finally substituting again

$$|G| = |Z(G)| + \sum_{i=i}^{n} [G : C(x)]$$

15. Centers of p-groups. Prove that every finite p-group has nontrivial center.

Let G be a finite group such that.

$$|G| = p^n$$

Recall the class equations

$$|G| = |Z(G)| + \sum_{i=i}^{n} [G : C(x)]$$

$$|\text{Cl}(x)| = [G : C(x)] = \frac{|G|}{|C(x)|}$$

Thus, $|C(x)|$ divides p^n. Thus, for some $k < n$

$$|C(x)| = p^k$$

Hence

$$[G : C(x)] = p^{n-k}$$

Since p divides $|G|$ and $|p|$ divides $[G : C(x)]$, then by our class equation, p must also divide $|Z(G)|$. Since $e \in Z(G)$ that means $Z(G) > 1$ and thus is non trivial. (Hence every group of order p^2 is abelian)

16. Cauchy's theorem. Prove that if a prime p divides $|G|$, then G contains an element of order p.

Suppose p divides the order of the G. Where $|G| = n$. Consider the set

$$X = \{(g_1, g_2, \cdots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$$

Since the first $p-1$ coordinates of an element in X can be chosen arbitrarily. This is because g_p is the inverse of the product of the first $p-1$ coordinates. Thus, $|X| = n^{p-1}$. Next we consider the cyclic group C_p acting on elements of X by cycling the coordinates. So for the generator $c \in C_p$ we get

$$c \cdot (g_1, g_2, \cdots, g_p) = (g_2, g_3, \cdots, g_1)$$

Fix $x \in X$. By the orbit stabilizer theorem

$$|C_p| = |\text{Orb}(x)||\text{Stab}(x)|$$

Since C_p has order p, then $|\text{Orb}(x)|$ is either p or 1. By the partition equation let r be the number of orbits with 1 elements and s be the number of orbits with p elements. We see that

$$n^{p-1} = r + sp$$

Since $p|n$, then $p|r$. We also know that $r > 0$ since the element $(e, e, ..., e) \in X$ has order 1. Thus, there are at least p orbits with one element. Let $a \neq e$ be such element. Then

$$(a, a, ..., a) \in X \implies a^p = e$$

Thus, $a \in G$ is an element of order p.

17. Sylow theorem application. Prove that every group of order 45 is abelian.

Let G be a group of order 45. By the 3rd Sylow theorem, the number of 3-Sylow subgroups is $n_3 \equiv 1 \mod 3$. However n_3 divides 45. Thus, $n_3 = 1$ and there exists only one normal subgroup of order 9. Similarly $n_5 \equiv 1 \mod p$. However n_5 divides 45. So $n_5 = 1$. Therefore only one normal subgroup of order 5 exists. Let H and K be normal subgroups of order 9 and 5 respectively. Thus, by Lagrange's theorem

$$|H \cap K| = 1 \implies |H \times K| = 45$$

Thus, $H \times K \cong G$ Every group of order 5 is abelian because all groups of prime order are cyclic and thus abelian. We also know that groups of order p^2 are also abelian and thus groups of order 9 are abelian. The direct product of two abelian groups is abelian. Thus, all groups of order 45 are abelian.

18. Sylow classification problem. Classify, up to isomorphism if possible, all groups of order pq, where $p < q$ are primes and $p \nmid q - 1$

Generally we want to show that a unique subgroup of a given order (meaning no other subgroups exist with that same order) is a normal subgroup. Let H be the only subgroup of G with order $|H|$. Consider an element $g \in G$. Given that the conjugate of a subgroup is a group we see that gHg^{-1} is a subgroup of G. Since left and right coset multiplications are bijective, then $|gHg^{-1}| = |H|$. But since H is the only group of order $|H|$, we see that $gHg^{-1} = H$ and thus

$$H \leq G, H \text{ has unique order} \implies H \trianglelefteq G$$

By Sylow, then $n_p \equiv 1 \mod p$ and $n_p \mid pq/p = q$. Thus, $n_p = q$ or $n_p = q$. But if $n_p = q$ then $n_p \mid q - 1$. A contradiction. Thus, $n_p = 1$. Next $n_q \equiv 1 \mod p$ and $n_q \mid pq/q = p$. But since $q > p$ then $n_q = 1$. Thus, there is only one unique group of each order. Let P be order p and Q be order q. By the Lagrange's theorem

$$P \cap Q = e$$

Then since there's only one of each order, P and Q are normal subgroups of the group of order pq we'll call G. Consider $g \in P$ and $h \in Q$. Then consider $x \in G$ to be $x = ghg^{-1}h^{-1}$ Since Q is normal, then $ghg^{-1} \in Q$. By closure $(ghg^{-1})h^{-1} \in Q$. Similarly Since P is normal, then $hg^{-1}h \in P$. By closure $g(hg^{-1}h^{-1}) \in P$. But if $x \in P \cap Q$ then $x = e$. Thus,

$$ghg^{-1}h^{-1} = e \implies gh = hg$$

Meaning elements of P and Q commute. Finally since P and Q are both prime orders. They are thus cyclic groups with

$$\langle g \rangle = P \quad \langle h \rangle = Q$$

Consider the group generated by $\langle gh \rangle$, we see that the order of ab is

$$\text{lcm}(pq) = pq = |G|$$

Since G is generated by the element ab it must be cyclic. Thus all groups of order pq with the given conditions is the cyclic group.

19. Direct products. State and prove the criterion for when a group G is isomorphic to an internal direct product $H \times K$

Suppose H and K are subgroups of G, and $HK = \{hk : h \in H, k \in K\}$. Then

$$H \trianglelefteq G,\ K \trianglelefteq G,\ H \cap K = \{e\},\ HK = G \implies H \times K \cong G$$

By the previous problem, we see that for all $k \in K$ and $h \in H$, then

$$hk = kh$$

Suppose that the representation of an element g is not unique. Then there exists h_1, h_2, k_1, k_2 such that

$$h_1k_1 = g = k_1k_2$$

then

$$\begin{aligned} h_1k_1 = g = h_2k_2 &\implies h_1k_1 = h_2k_2 \\ &\implies (h_2^{-1})h_1k_1(k_1^{-1}) = (h_2^{-1})h_2k_2(k_1^{-1}) \\ &\implies h_2^{-1}h_1 = k_2k_1^{-1} \\ &\implies h_2^{-1}h_1 = e = k_2k_1^{-1} \quad \text{since } H \cap K = \{e\} \\ &\implies h_1 = h_2,\ k_1 = k_2 \end{aligned}$$

Thus $hk = g$ is unique. Consider the the map $\varphi : H \times K \to G$ by

$$\varphi(h, k) = hk$$

Since $g = hk$ is unique the map is both well defined and surjective. Next evaluate $\varphi(h_1, k_1)\varphi(h_2, k_2)$

$$\begin{aligned} \varphi(h_1, k_1)\varphi(h_2, k_2) &= h_1k_1h_2k_2 \\ &= h_1h_2k_1k_2 \\ &= \varphi(h_1h_2, k_1k_2) \end{aligned}$$

Hence φ is also a homomorphism. Finally calculate $\ker \varphi$.

$$\begin{aligned} \varphi(h, k) = e &\implies hk = e \\ &\implies k = h^{-1} \\ &\implies h = e, k = e. \qquad \text{Since } H \cap K = \{e\} \end{aligned}$$

Since $\ker \varphi = \{e, e\}$ is trivial, then φ is injective. Thus φ is a bijective homomorphism. Thus,

$$G \times K \cong G$$

20. Semidirect products. Define $N \rtimes_\varphi H$. Classify all semidirect products $C_7 \rtimes C_3$

Let N and H be groups. Consider the homomorphism $\varphi : H \to \text{Aut}(N)$ by

$$\varphi(h_1) = \varphi_{h_1}$$

Thus the semi direct product is defined as all pairs (n, h) where multiplication is defined by

$$N \rtimes_\varphi H : (n_1, h_1)(n_2, h_2) = (n_1\varphi_{h_1}(n_2), h_1h_2)$$

Consider all homomorphisms $\varphi : C_3 \to \text{Aut}(C_7)$. Recall that

$$\text{Aut}(C_7) = \{\sigma_n : C_7 \to C_7 : \sigma \text{ is an isomorphism}\}$$

Where $\sigma_n(g) = g^k$. Also recall

$$\text{Aut}(C_p) \cong C_{p-1} \quad \text{Aut}(C_7) \cong C_6$$

Define

$$C_3 = \langle a : a^3 = e\rangle \quad C_6 = \langle b : b^6 = e\rangle$$

Thus all homomorphisms from $C_3 \to C_6$ send the generator of C_3, being a, to elements whose order divide 3 in C_6, thus the set of homomorphisms is the following.

$$\varphi(a) = e, \varphi(a) = b^2, \varphi(a) = b^4$$

Finally converting back into terms of $\text{Aut}(C_7)$ we get the following homomorphisms

$$\varphi(a) = \sigma_1, \varphi(a) = \sigma_2, \varphi(a) = \sigma_4$$

Thus we get the following groups. For $\varphi(a) = \sigma_1$ we have the identity automorphism, meaning it's the direct product

$$C_7 \rtimes_\varphi C_3 \cong C_7 \times C_3 \cong C_{21}$$

For $\varphi(a) = \sigma_2$

$$C_7 \rtimes_\varphi C_3 \cong G_1$$

Where G_1 is defined as

$$G_1 \langle a, x : x^7 = e, a^3 = e, ax = x^2a\rangle$$

Finally for $\varphi(a) = \sigma_2$,

$$C_7 \rtimes_\varphi C_3 \cong G_2$$

Where G_2 is defined as

$$G_2 \langle a, x : x^7 = e, a^3 = e, ax = x^4a\rangle$$

21. Finitely generated abelian groups. State the classification theorem for finitely generated abelian groups. Decompose:

$$\mathbb{Z}/72\mathbb{Z} \oplus \mathbb{Z}/180\mathbb{Z}$$

Let G be a finitely generated abelian group. There exists integers

$$r \geq 0, s \geq 0, 2 \leq d_1 \mid d_2 \mid \cdots \mid d_s$$

such that

$$G \cong \mathbb{Z}^r \oplus \mathbb{Z}_{d_1} \oplus \mathbb{Z}_{d_2} \oplus \cdots \oplus \mathbb{Z}_{d_s}$$

The primary decomposition tool is for integers m, n then

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \oplus \mathbb{Z}_n \iff \gcd(m, n) = 1$$

Decomposing the above, we see that

$$72 = 8 \times 9 \quad 180 = 4 \times 5 \times 9$$

$$\begin{aligned}\mathbb{Z}/72\mathbb{Z} \oplus \mathbb{Z}/180 &\cong \mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/9 \oplus \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/9\mathbb{Z} \\ &\cong (\mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/9) \oplus (\mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/9\mathbb{Z}) \\ &\cong \mathbb{Z}/36\mathbb{Z} \oplus \mathbb{Z}/360\end{aligned}$$

22. Ring definitions. Define ring, integral domain, field, ideal, quotient ring, prime ideal, and maximal ideal. Give examples showing the implications are not reversible where appropriate.

A ring $(R, +, \cdot)$ is a set R with two operations $+, \cdot$ with the following axioms.

Abelian group under addition with additive identity 0

$$\begin{aligned} \text{Associative } & \forall a, b, c \in R, (a+b)+c = a+(b+c) \\ \text{Commutative } & \forall a, b \in R, a+b = b+a \\ \text{Identity } & \exists 0 \in R, a+0 = a \\ \text{Inverses } & \forall a \in R, \exists a^{-1} \in R, a+a^{-1} = 0 \end{aligned}$$

Monoid under multiplication with multiplicative identity 1 and distributive properties

$$\begin{aligned} \text{Associative } & \forall a, b, c \in R, (a \cdot b) \cdot c = a \cdot (b \cdot c) \\ \text{Identity } & \exists 1 \in R, a \cdot 1 = 1 \cdot a = a \\ \text{Right Distributivity } & \forall a, b, c \in R, (a+b) \cdot c = a \cdot c + b \cdot c \\ \text{Left Distributivity } & \forall a, b, c \in R, a \cdot (b+c) = a \cdot b + a \cdot c \end{aligned}$$

An integral domain is a nonzero commutative ring with no nonzero zero divisors. That being, for all $a, b \in R$ where $a, b \neq 0$ then $ab \neq 0$. A counter example would be the ring $\mathbb{Z}/6\mathbb{Z}$ where $a = 3$ and $b = 2$ but $ab = 0$.

A field $(F, +, \cdot)$ is a Ring with multiplicative inverses and commutative multiplication. That being

$$\begin{aligned} \text{Commutative } & \forall a, b \in F, a \cdot b = b \cdot a \\ \text{Inverses } & \forall a \in F, \exists a^{-1} \in F, a \cdot a^{-1} = 1 \end{aligned}$$

An ideal $(I, +)$ is a subgroup of the additive group R. Where for all $a \in R$ and all $x, y \in I$ then

$$x + y \in I \quad -x \in I \quad rx \in I \text{ (for a left ideal)} \quad xr \in I \text{ (for a right ideal)}$$

Quotient ring: For an ideal $I \subset R$ where $I \neq R$. Then a quotient ring R/I is defined as

$$R/I = \{r + I : r \in R\}$$

With addition and multiplication defined by

$$(a+I)+(b+I) = (a+b)+I \quad (a+I)(b+I) = (ab+I)$$

Let P be a Ideal of R where $P \neq R$. Then P is a prime ideal if for all $ab \in P$ then $a \in P$ or $b \in P$. Similarly P is an ideal if and only if R/P is an integral domain.

Let M be a Ideal of R where $M \neq R$. Then M is a maximal ideal if for any ideal J with $M \subseteq J$ then

$$M = J \quad \text{or} \quad J = R$$

Similarly $M \subset R$ is maximal ideal if and only if R/M is a field. And thus a maximal ideal is a prime ideal. The converse however isn't true. Take the ring $\mathbb{Z}$. The element $\{0\}$ is prime because

$$\mathbb{Z}/\{0\} \cong \mathbb{Z}$$

However any set $p\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

23. Ring homomorphisms. Prove the first isomorphism theorem for rings.

Let $\varphi : R \to S$ be a ring homomorphism and $\ker\varphi$ denotes the kernel of φ. Then we want to show that

$$R/\ker\varphi \cong \varphi(R)$$

Define the kernel such that

$$\ker\varphi = \{r \in R : \varphi(r) = 0_S\}$$

Shorten $\ker\varphi = K$ for brevity. Since φ is a ring homomorphism $0_R \in K$. Take $x, y \in K$ then

$$\varphi(x+y) = \varphi(x) + \varphi(y) = 0_S + 0_S$$

$$\varphi(x-y) = \varphi(x) - \varphi(y) = 0_S - 0_S$$

Meaning K is closed under addition and subtraction. Finally take $r \in R$ and $k \in K$. Evaluating $\varphi(rk)$

$$\varphi(rk) = \varphi(r)\varphi(k) = \varphi(r)0_S = 0_S$$

Thus K is an ideal of R. We also define R/K as

$$R/K = \{r + K : r \in R\}$$

Next consider the function $\psi : R/K \to \varphi(R)$ such that

$$\psi(r+K) = \varphi(r)$$

By the first isomorphism theorem for groups ψ is well defined and an isomorphism for $(R, +)$. Thus it would suffice to show that ψ holds the multiplicative property for rings. Evaluating $\psi(x+K)\psi(y+K)$

$$\begin{aligned}\psi(x+K)\psi(y+K) &= \varphi(x)\varphi(y) \\ &= \varphi(xy) \qquad \text{by ring homomorphism properties} \\ &= \psi(xy+K) \qquad \text{by the defined map}\end{aligned}$$

Thus ψ is a ring isomorphism and thus

$$R/K \cong \varphi(R)$$

24. Ideals in $\mathbb{Z}$. Prove every ideal of $\mathbb{Z}$ is principal. Then describe all quotient rings of $\mathbb{Z}$.

Recall that a principal ideal in a commutative ring is an ideal generated by a single element $\langle a\rangle$ defined as

$$\langle a\rangle = \{ra : r \in R\}$$

We see that we get a family of ideals for all $n \in \mathbb{Z}$ were $\langle n\rangle = n\mathbb{Z}$ is an ideal. Suppose there existed an ideal $I \subset \mathbb{Z}$ where I is not a principal ideal. I can't be a trivial ideal because $I \neq \langle 0\rangle$. Then for all $a \in \mathbb{Z}$, we see that

$$\langle a\rangle \subseteq I$$

But since I is not the principal ideal, there exists $b \in I \subset \mathbb{Z}$ such that $b \notin \langle a\rangle$. I should contain all multiples of b as well as every linear combination of b and a. Thus

$$\langle \text{lcm}(a,b)\rangle \subseteq I$$

Again however there has to exist $c \in I$ that isn't generated by $\text{lcm}(a,b)$. But that in tern would provide a new principal ideal. Thus I must be principal. All quotient rings of $\mathbb{Z}$ are in the form of the following

$$\mathbb{Z}/n\mathbb{Z}$$

It's just all the n-modular systems. Notice that for a prime p we see $\mathbb{Z}/p\mathbb{Z}$ there are no zero divisors. Conversely, $\mathbb{Z}/6\mathbb{Z}$ has zero divisors 2 and 3 since.

$$2 \cdot 3 \equiv 0 \mod 6$$

Hence $p\mathbb{Z}$ is a prime ideal

25. Chinese remainder theorem. State and prove the Chinese remainder theorem. Use it to compute:

$$\mathbb{Z}/360\mathbb{Z}$$

as a product of rings modulo prime powers.

For a system of modular equations with one unknown x in the form of

$$x \equiv a_1 \mod n_1, \quad x \equiv a_2 \mod n_2, \quad \cdots \quad x \equiv a_k \mod n_k$$

Where n_i and n_j are pairwise coprime. Then there exists unique a such that all solutions x are

$$x \equiv a \mod N$$

where $N = n_1 \cdot n_2 \cdot ... \cdot n_k$

Proof (existence): Denote N_i as $\frac{N}{n_i}$. Then we see that for our solution x and all $i \in \{1, 2, ..., k\}$

$$N_i \equiv 1 \mod n_i$$

Since all n_j are pairwise coprime to n_i for $i \neq j$ and N_i is the product of every n_j. Since $N_i x_k \equiv a_i \mod n_i$ then there exists unique x_k such that

$$N_i x_k \equiv a_i \mod n_i$$

Consider the number x_0 such that

$$x_0 = \sum_{i=1}^{k} N_i x_i$$

Notice for each $i, j \in \{1, 2, ..., n\}$ we see

$$\begin{aligned} n_i \mid N &\implies n_i \mid N_j n_j \\ &\implies n_i \mid N_j \\ &\implies n_i \mid N_j x_j \\ &\implies N_j x_j \equiv 0 \mod n_i \\ &\implies x_0 = \sum_{i=1}^{k} N_i x_i \equiv a_n \mod n_i \end{aligned}$$

Thus x_0 is a solution to the system of modular equations.

Proof (uniqueness): Suppose there is another solution x' such that for all $i \in \{1, 2, ..., n\}$

$$x_0 \equiv x' \equiv a_i \mod n_i$$

Then we see

$$\begin{aligned} x_0 \equiv x' \equiv a_i \mod n_i &\implies x_0 \equiv x' \mod n_i \\ &\implies n_i \mid x_0 - x' \\ &\implies N \mid x_0 - x' \\ &\implies x_0 \equiv x' \mod N \end{aligned}$$

Next we want to show that

$$\mathbb{Z}/N\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

Where n_i and n_j are pairwise coprime. N is the product of all n_i.

Consider the map

$$\varphi : \mathbb{Z} \to \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

By

$$\varphi(x) = (x \mod n_1, x \mod n_2, ..., x \mod n_k)$$

We will shorten to $(a_1, a_2, ..., a_n)$ for brevity. Note that the map is surjective and well defined because mappings residue classes are well defined. Next suppose that $\varphi(x) = \varphi(y)$ then we see that

$$\begin{aligned} \varphi(x) = \varphi(y) &\implies (a_1, a_2, ..., a_n) = (b_1, b_2, ..., b_n) \\ &\implies n_i \mid x - y \\ &\implies x \equiv y \mod N \text{ by the (CRT)} \end{aligned}$$

Meaning there exists a bijective map Φ such that

$$\Phi : \mathbb{Z} \mod N = \mathbb{Z}/N\mathbb{Z} \to \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

We also see that Φ is a ring homomorphism as addition, subtraction, and multiplication are preserved. Thus Φ is a ring isomorphism. Thus we can decompose $\mathbb{Z}/360\mathbb{Z}$ into rings with prime powers. We see that $360 = 2^3 \times 3^2 \times 5$ and thus

$$\mathbb{Z}/360\mathbb{Z} \cong \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$$

26. PIDs and UFDs. Define Euclidean domain, PID, and UFD. Prove:

$$\textbf{Euclidean domain} \implies \textbf{PID} \implies \textbf{UFD}$$

Euclidean Domain: An integral domain R equipped with a norm function $f : R/\{0\} \to \mathbb{N}_0$ mapping non-zero elements to non-negative integers and includes division with remainder. That is for $a, b \in R$ where $b \neq 0$ then there exists q, r such that

$$a = bq + r$$

Where $r = 0$ or $f(r) < f(b)$.

Principal ideal domain (PID): An integral domain R in which every ideal I is generated by a single element. Meaning all ideals are of the form $aR = \{ar : r \in R\}$ for some $a \in R$.

Unique Factorization Domain (UFD): An integral domain in which ever non-zero, non-unital element can be factored uniquely into irreducible elements. Meaning for all $x \in R$ where $x \notin 0, 1$ then there exist irreducible elements p_i such that

$$x = p_1 \cdot p_2 \cdot ... \cdot p_n$$

is a unique factorization and elements that differ between factorizations will be associates.

Let D be a PID and . Thus every ideal $I \subset D$ is of the form $aD = \langle a \rangle$ for some $a \in D$. Consider a non-zero, non-unital element $x \in D$.

First we want to show that **Euclidean Domain $\implies$ PID**
Let D be a Euclidean Domain with a norm function $N : D \to \mathbb{N}$ and let I be an arbitrary ideal of D. Choose $b \in I$ such that $N(b)$ is minimal. Thus the ideal $\langle b \rangle \subseteq I$. Next choose arbitrary $a \in I$. By D being a

Euclidean domain we get that there exists $q, r \in D$ such that $a = bq + r$ where $N(r) < N(b)$ or $r = 0$. Since $a \in I$ and $bq \in \langle b \rangle \subseteq I$. Then we see that $r = a - bq \in I$. But $N(r) < N(b)$ contradicts b being the minimal element. Thus $r = 0$ and $a = bq$ for some $q \in D$. Hence for any $a \in I \implies a \in \langle b \rangle$. Thus $I \subseteq \langle b \rangle$ and hence

$$I = \langle b \rangle$$

Hence for any ideal I in a euclidean domain, it's a principal ideal. Thus D is a PID.

Next we want to show that **PID $\implies$ UFD**
Let D be a PID and $a, b \in D$. Suppose $\langle b \rangle \subseteq \langle a \rangle$. Then $b \in \langle a \rangle \implies b = aq$ for some $q \in D$. Then $a \mid b$. So we showed that $a \mid b \implies \langle b \rangle \subseteq \langle a \rangle$. Similarly, if $a \mid b$, then $b = aq$ for some $q \in D$. If $x \in \langle b \rangle$ then $x = by$ and thus $x = (aq)y$ or $x = a(qy)$ meaning $x \in \langle a \rangle$. Thus

$$a \mid b \iff \langle b \rangle \subseteq \langle a \rangle$$

Suppose $a \in D$ that is non-zero and non-unit. Write $a = a_1 b_1$ where a_1 is not a unit. Then write $a_1 = a_2 b_2$ where a_2 is not a unit. Continuing this we write $a_k = a_{k+1} b_{k+1}$ where a_{k+1} is not a unit. Meaning

$$a_1 \mid a, a_2 \mid a_1, ..., a_{k+1} \mid a_k$$

Recall that $a \mid b \iff \langle b \rangle \subseteq \langle a \rangle$. Thus we get that

$$\langle a \rangle \subseteq \langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \langle a_3 \rangle \subseteq ...$$

Since D satisfies the acceding chain condition (because a PID is a Noetherian Ring), then there exists N such that for all $n \geq N$ then $\langle a_N \rangle = \langle a_n \rangle$. Meaning a_N and a_n are associates for all $n \geq N$. Thus the only way to factor a_N is by $a_N = pu$ where p is irreducible and u being a unit. Thus for all $a \in D$ we can factor out an irreducible $a = p_1 x_1$ where x is something else. Repeating this process we can write $x_1 = p_2 x_2$ where p_2 is irreducible, $x_2 = p_3 x_3$ where p_3 is irreducible, $x_k = p_{k+1} x_{k+1}$ where p_{k+1} is irreducible. We can create an ideal chain like the following

$$\langle x_1 \rangle \subseteq \langle x_2 \rangle \subseteq \langle x_3 \rangle \subseteq ...$$

Thus we have a terminating sequence starting with $\langle x_N \rangle = \langle x_{N+1} \rangle ...$ We then see that x_N is irreducible. We then see that $x_{N-1} = p_N x_N$. Thus x_{N-1} is a product of irreducibles. Set $x_N = p_{N+1}$. Thus we can write

$$a = p_1 x_1 = p_1 p_2 x_2 = p_1 p_2 p_3 ... x_N = p_1 p_2 p_3 ... p_{N+1}$$

Thus we took a non-zero unit and proved we can factor it into a product of irreducibles. Now we need to prove uniqueness of each factorization. Suppose there were two factorizations of some non-zero element $a \in D$

$$a = p_1 p_2 p_3 ... p_r = q_1 q_2 q_3 ... q_s$$

Thus we want to show that $r = s$ and $p_i = u_j q_j$ for some permutation if the indexing.

Assume that $r \neq s$ and WLOG assume $r < s$. We know that $p_i \mid a$ and thus $p_1 \mid q_1 q_2 ... q_s$. Since p_1 is prime then there exists q_j such that $p_1 \mid q_j$. We will re-index the product such that $p_1 \mid q_1$. Thus $q_1 = u_1 p_1$. Substituting in for q_1 we see

$$p_1(p_2 p_3 ... p_r) = u_1 p_1 (q_2 q_3 ... q_s)$$

Canceling p_1

$$p_2 p_3 ... p_r = u_1 q_2 q_3 ... q_s$$

Completing the same process for p_2 we see $q_2 = u_2 p_2$ and substituting again

$$p_2(p_3 ... p_r) = u_1 u_2 p_2 (q_3 ... q_s)$$

Repeat until we cancel all such p_i. We took such that $r < s$ so the LHS will cancel before the RHS. We will then get

$$1 = u_1 u_2 ... u_r q_{r+1} ... q_s$$

Thus we see that for all q_j for $j \in \{r+1, ..., s\}$ are units. But units are not irreducibles. A contradiction. Thus $r = s$. And by the same argument we also showed that $p_i = u_i q_i$ for some re-indexing of i. Hence factorizations of all non-zero elements of a PID are unique up to associates. By definition we proved that

Every PID is a UID and every Euclidean Domain is a PID

27. Polynomial irreducibility. Determine whether the following polynomial is irreducible in $\mathbb{Q}$

$$f(x) = x^4 + 4x^3 + 6x^2 + 2x + 1$$

Suppose that the polynomial is reducible. Then we have $a(x), b(x) \in \mathbb{Q}[x]$

$$f(x) = a(x)b(x)$$

Where $0 < \deg a(x), b(x) < f(x)$. Similarly if we choose $c \in \mathbb{Q}$ then the map $x \to y + c$ must also keep the polynomial reducible as shown here

$$f(y + c) = a(y + c)b(y + c)$$

Notice that $f(x) = (x + 1)^4 - 2x$. Mapping $x \to y - 1$ we see

$$\begin{aligned} f(x) &= (x + 1)^4 - 2x \\ f(x) &= (y - 1 + 1)^4 - 2(y - 1) \\ f(x) &= y^4 - 2y + 2 \end{aligned}$$

Note by Eisenstein's Criterion, we see that for $p = 2$ our polynomial is irreducible in $\mathbb{Q}$.

28. Modules over a PID. State the structure theorem for finitely generated modules over a PID. Use it to classify finitely generated abelian groups.

Let R be a PID, and let M be an $R-$Module. Then for some $n \geq 0$ we have

$$M \cong R^n \oplus \text{Tor}(M)$$

Note that M is an abelian group under addition and scalar multiplication

$$R \times M \to M \qquad (r, m) \mapsto rm$$

Conversely, let G be an abelian group. Let $g \in G$ and $n \in \mathbb{Z}$. Define

$$g^n \begin{cases} g \cdot g \cdot ... \cdot g & n > 0 \\ e & n = 0 \\ g^{-(-n)} & n < 0 \end{cases}$$

By this definition G is a $\mathbb{Z}-$module. Take G to be finitely generated.

$$G = \langle g_1, g_2 ..., g_n \rangle$$

Hence for all $g \in G$ we can write it as

$$g = g_n^{a_n} \cdot g_n^{a_n} ... g_n^{a_n} \qquad a_i \in \mathbb{Z}$$

Which gives a finitely generated $\mathbb{Z}$-module. Hence finitely generated $\mathbb{Z}$-modules are are finitely generated abelian groups and the converse holds. Next consider for some finitely generated group G

$$\text{Tor}(G) = \{g \in G \mid \exists 0 \neq n \in \mathbb{Z}, g^n = e\}$$

We see that

$$g \in \text{Tor}(G) \iff g \text{ has finite order}$$

Since $\mathbb{Z}$ is a PID, then for all $0 \neq n \in \mathbb{Z}$ we see that $\mathbb{Z}/(n)$ is a cyclic $\mathbb{Z}$-module. Hence a cyclic group. By the invariant factor decomposition

$$\text{Tor}(M) \cong R/(a_1) \oplus R/(a_2) \oplus ... \oplus R/(a_k) \qquad a_1 \mid a_2 \mid ... \mid a_k$$

Applying our definition

$$\text{Tor}(G) \cong \mathbb{Z}/(a_1) \oplus \mathbb{Z}/(a_2) \oplus ... \oplus \mathbb{Z}/(a_k) \qquad a_1 \mid a_2 \mid ... \mid a_k$$

$$\text{Tor}(G) \cong \bigoplus_{i=1}^{k} \mathbb{Z}/(a_i) \qquad a_1 \mid a_2 \mid ... \mid a_k$$

We will write the cyclic groups of orders k as $\mathbb{Z}_k$ from now on. Hence our definition becomes for a finitely generated group G we get

$$G \cong \mathbb{Z}^n \oplus \bigoplus_{i=1}^{k} \mathbb{Z}_{a_i} \qquad a_i \mid a_{i+1}$$

Hence we regained the fundamental theorem for finitely generated abelian groups. And by that theorem, any finitely generated abelian groups G, all a_i are uniquely determined. We proved existence and uniqueness. Thus we classified all finitely generated abelian groups.

29. Field extensions. Let $\alpha = \sqrt[3]{2}$. Compute:

$$[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}]$$

Where ω is a primitive cube root of unity. Identify the splitting field of $x^3 - 2$ over $\mathbb{Q}$

By the tower rule we see

$$[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}] = [\mathbb{Q}(\alpha, \omega) : \mathbb{Q}(\alpha)] \cdot [\mathbb{Q}(\alpha) : \mathbb{Q}]$$

Computing $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ the minimal polynomial $\alpha^3 - 2$ is degree 3 (as it satisfies Eisenstein's criterion for $p = 3$). Next we see that $\omega \notin \mathbb{Q}(\alpha)$. Hence we can adjoin ω with no issue. The minimal polynomial $\omega^2 + \omega + 1$ is degree 2. Hence $[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}(\alpha)] = 2$ Thus

$$[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}] = [\mathbb{Q}(\alpha, \omega) : \mathbb{Q}(\alpha)] \cdot [\mathbb{Q}(\alpha) : \mathbb{Q}] = 2 \cdot 3 = 6$$

30. Galois theory. State the fundamental theorem of Galois theory. Then compute the Galois group of $x^3 - 2$ over $\mathbb{Q}$, list all intermediate fields of its splitting field, and match them with subgroups of the Galois group.

The fundamental theorem of Galois theory states that any subgroup of the Galois group corresponds to intermediate field extensions. Our Galois group $G \cong S_3$ and thus our table looks like this.

Subgroup of $\text{Gal}(K/\mathbb{Q})$	Order	Index	Fixed Intermediate Field	Degree over $\mathbb{Q}$
$G \cong S_3 = \langle \sigma, \tau \rangle$	6	1	$\mathbb{Q}$	1
$\langle \sigma \rangle = \{1, \sigma, \sigma^2\}$	3	2	$\mathbb{Q}(\omega)$	2
$\langle \tau \rangle = \{1, \tau\}$	2	3	$\mathbb{Q}(\sqrt[3]{2})$	3
$\langle \sigma^2\tau \rangle = \{1, \sigma^2\tau\}$	2	3	$\mathbb{Q}(\sqrt[3]{2}\,\omega)$	3
$\langle \sigma\tau \rangle = \{1, \sigma\tau\}$	2	3	$\mathbb{Q}(\sqrt[3]{2}\,\omega^2)$	3
$\{1\}$	1	6	$\mathbb{Q}(\sqrt[3]{2}, \omega)$	6